

ST10498624-CLDV7112-Project2

Student Number: ST10498624

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Module Code: CLDV7112

Module Name: Cloud Development B

Assessment: Project 2

GitHub Repository: <https://github.com/iamgeraldlehana/CLDV7112w-Project-2>

Function App URL: <https://gmlretailfunctions.azurewebsites.net>

Function Endpoints:

POST <https://gmlretailfunctions.azurewebsites.net/api/store-customer>

POST <https://gmlretailfunctions.azurewebsites.net/api/enqueue-order>

POST <https://gmlretailfunctions.azurewebsites.net/api/upload-blob>

POST <https://gmlretailfunctions.azurewebsites.net/api/upload-log>

Part A: Azure Functions

Four HTTP-triggered Azure Functions were created, each integrating with a different Azure Storage service. All functions read the storage connection string from the `AzureStorage` environment variable and create resources automatically using `CreateIfNotExistsAsync()`.

Function 1 — StoreCustomer (Azure Table Storage)

Route: POST `/api/store-customer`

Stores customer profiles in Azure Table Storage (table: `customers`). Accepts `FullName`, `Email`, `Phone`, and `Address`. Returns the saved entity as JSON.

```
Last login: Sat May  9 20:25:13 on ttys008
❯ curl -X POST "http://localhost:7071/api/store-customer" -H "Content-Type: application/json" -d '{"FullName":"Gerald Lehana","Email":"gerald@test.com","Phone":"0611050573","Address":"Randfontein"}'
{"PartitionKey":"Customer","RowKey":"f183bdfd-628f-4fe5-bd39-fa26618f35ca","Timestamp":null,"ETag":null,"FullName":"Gerald Lehana","Email":"gerald@test.com","Phone":"0611050573","Address":"Randfontein"}
❯ curl -X POST "https://gmlretailfunctions.azurewebsites.net/api/store-customer?code=S3sCBJNVTC4aqHb8dm5du5m1Ac_u8CGPt4SzzXz3jg0lAzFu56EIEw" -H "Content-Type: application/json" -d '{"FullName":"Gerald Lehana","Email":"gerald@test.com","Phone":"0611050573","Address":"Randfontein"}'
at 20:26:25
```

```
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Home > App Services > GMLRetailFunctions
StoreCustomer | Code + Test
GMLRetailFunctions

Code + Test Integration Function Keys Invocations Logs Metrics
Save Discard Refresh Test/Run Get function URL Disable Delete Upload Resource JSON Send us your feedback

Editing .NET isolated Function Apps is not supported in the Azure portal. Use your local development environment to edit this Function App.

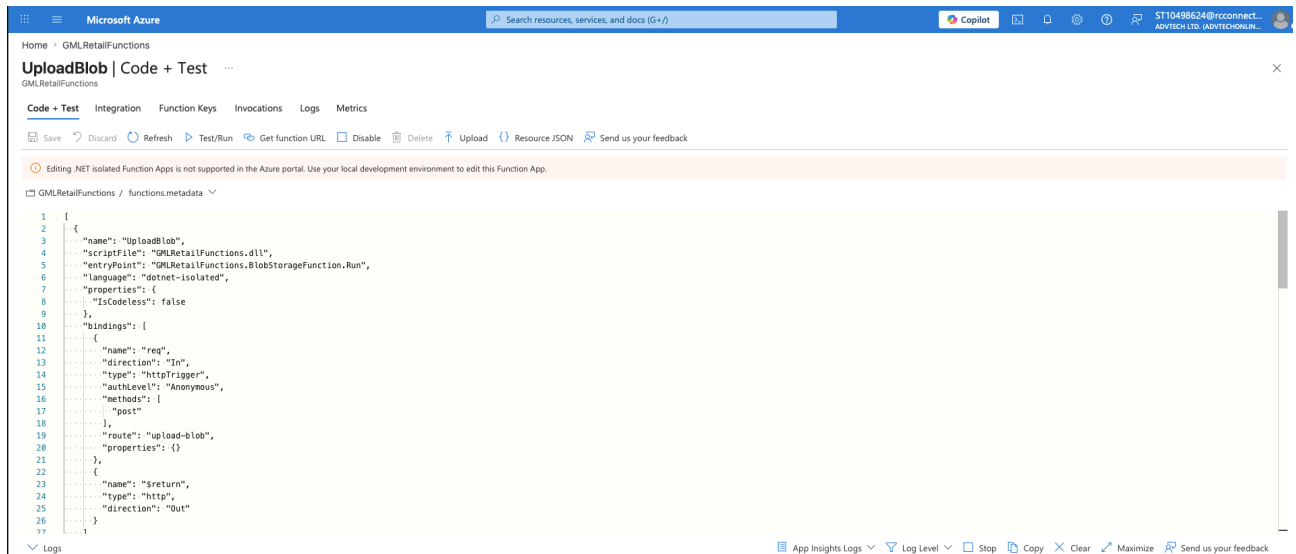
GMLRetailFunctions / functions.metadata
1 {
2   "name": "uploadBlob",
3   "scriptFile": "GMLRetailFunctions.dll",
4   "entryPoint": "GMLRetailFunctions.BlobStorageFunction.Run",
5   "language": "dotnet-isolated",
6   "properties": {
7     "isCodeless": false
8   },
9   "bindings": [
10    {
11      "name": "req",
12      "direction": "In",
13      "type": "httpTrigger",
14      "authLevel": "Anonymous",
15      "methods": [
16        "post"
17      ],
18      "route": "upload-blob",
19      "properties": {}
20    },
21    {
22      "name": "return",
23      "type": "http",
24      "direction": "Out"
25    }
26  ]
27 }
```

Function 2 — UploadBlob (Azure Blob Storage)

Route: POST /api/upload-blob

Uploads base64-encoded files to Azure Blob Storage (container: mediafiles). Returns the blob name on success.

```
Last login: Sat May 9 20:27:22 on ttys008
~ % curl -X POST "http://localhost:7071/api/upload-blob" -H "Content-Type: application/json" -d '{"FileName":"product_001.jpg","ContentBase64":"iVBORw0KGgoAAAANSUhEUgAAAAEAAAABCAyAAAFcSJAADU1EQVR42mNk+M9QDwADhgGAWjR9awAAAAABJRUSErkJggg=="}'
{"blobName":"product_001.jpg"}
~ % curl -X POST "https://gmlretailfunctions.azurewebsites.net/api/upload-blob?code=S3sCBJNVTC4aqHb8dm5du5m1Ac_u8CGPt4SzzXz3jg01AzFu56EIEw" -H "Content-Type: application/json" -d '{"FileName":"product_001.jpg","ContentBase64":"iVBORw0KGgoAAAANSUhEUgAAAAEAAAABCAyAAAFcSJAADU1EQVR42mNk+M9QDwADhgGAWjR9awAAAAABJRUSErkJggg=="}'
```

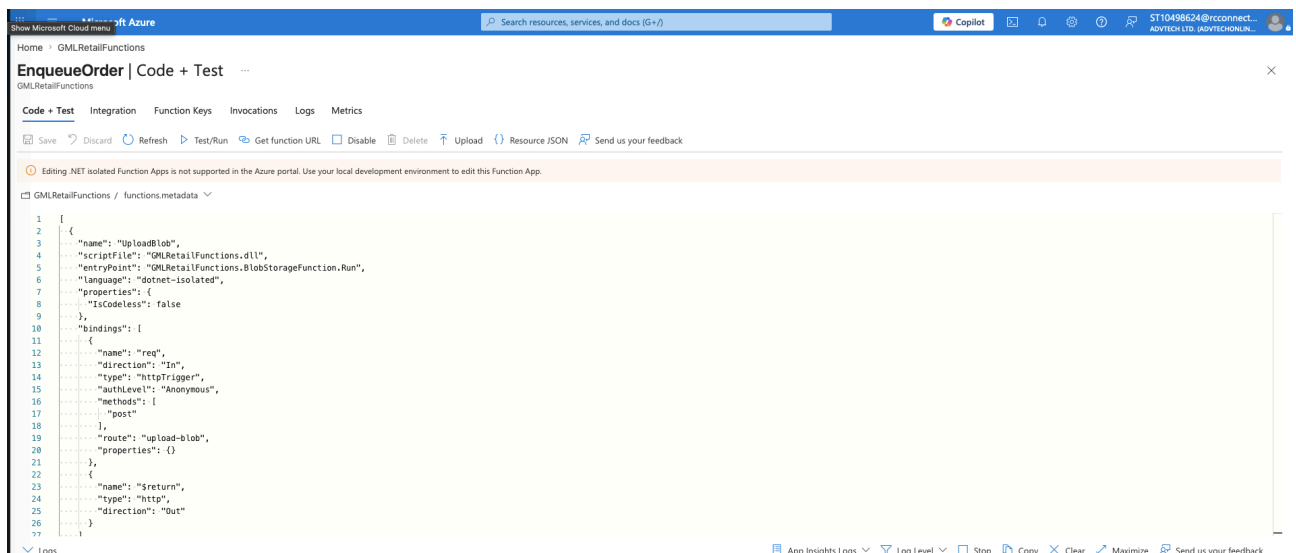


Function 3 — EnqueueOrder (Azure Queue Storage)

Route: POST /api/enqueue-order

Sends order messages to Azure Queue Storage (queue: orders). Returns confirmation with the enqueued message.

```
Last login: Sat May 9 20:26:05 on ttys007
~ % curl -X POST "http://localhost:7071/api/enqueue-order" -H "Content-Type: application/json" -d '{"Message":"Processing order #1001 for Gerald Lehana"}'
{"status":"enqueued","message":"Processing order #1001 for Gerald Lehana"}
~ % curl -X POST "https://gmlretailfunctions.azurewebsites.net/api/enqueue-order?code=S3sCBJNVTC4aqHb8dm5du5m1Ac_u8CGPt4SzzXz3jg01AzFu56EIEw" -H "Content-Type: application/json" -d '{"Message":"Processing order #1001 for Gerald Lehana"}'
```

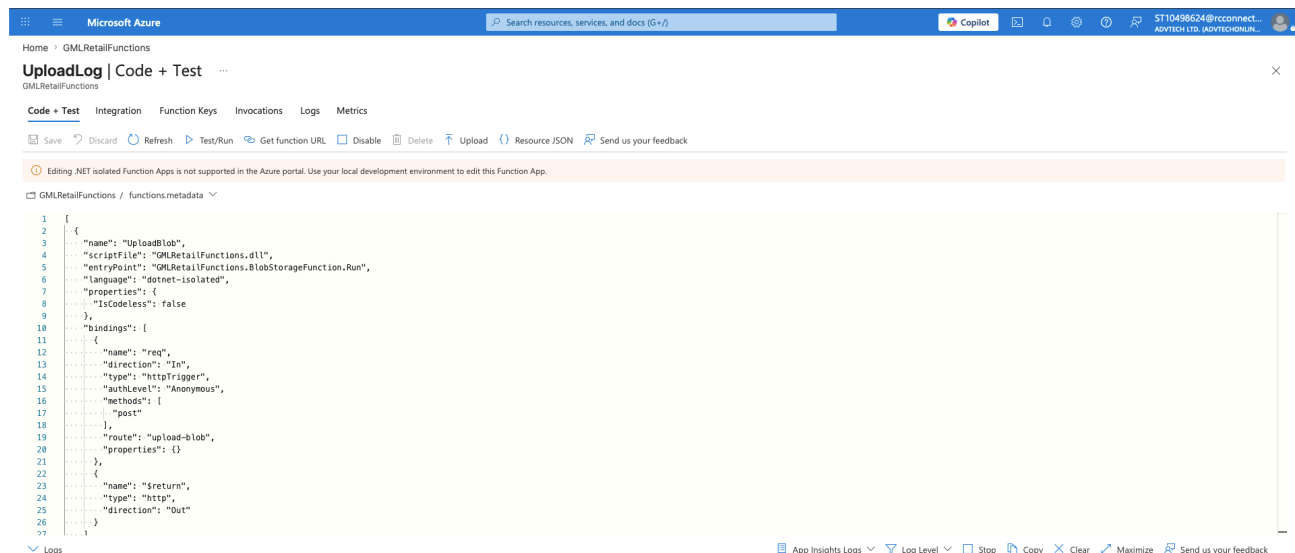


Function 4 — UploadLog (Azure File Storage)

Route: POST /api/upload-log

Uploads base64-encoded log files to Azure File Storage (share: logfiles). Returns confirmation with the file name.

```
Last login: Sat May 9 20:28:28 on ttys009
curl -X POST "http://localhost:7071/api/upload-log" -H "Content-Type: application/json" -d '{"FileName":"app_log_20260509.log","ContentBase64":"QXBwbGljYXRpb24gc3RhcmlZCBzdWVjZ0NzZnVsbHkuCg=="}'
{"status":"uploaded","fileName":"app_log_20260509.log"}
curl -X POST "https://gmlretailfunctions.azurewebsites.net/api/upload-log?code=S3sCBJNVTC4aqHb8dm5du5m1Ac_u8CGPt4SzzXz3jg0LAzFu56EIEw" -H "Content-Type: application/json" -d '{"FileName":"app_log_20260509.log","ContentBase64":"QXBwbGljYXRpb24gc3RhcmlZCBzdWVjZ0NzZnVsbHkuCg=="}'
at 20:29:29
```



Part B: Using Services to Improve Customer Experience

Azure Event Hubs

Description

Azure Event Hubs is a cloud-based data streaming service that can receive and process millions of events per second from different sources in real time.

Mechanism

Publishers like POS systems or web apps send events to Event Hubs. The events are stored in partitions and consumers read from them in order. Event Hubs uses a pull model so consumers read at their own pace.

How it adds value to end users

In Gerald's Retail, Event Hubs could stream customer activity like product views and clicks in real time. This allows the app to show personalised product recommendations instantly, improving the shopping experience.

Azure Service Bus

Description

Azure Service Bus is a cloud messaging service that allows different parts of an application to communicate reliably using queues and topics.

Mechanism

A sender puts a message into a queue or topic. Service Bus holds it until a receiver picks it up. Topics allow one message to be delivered to multiple subscribers at the same time.

How it adds value to end users

In Gerald's Retail, Service Bus could handle order confirmation messages. When a customer places an order, a message is sent to Service Bus which triggers notifications, inventory updates, and payment processing reliably without any messages being lost.